

Advanced (Habanero)

- Q1.** What is the purpose of a box-and-whisker plot?
(A) To show the mean of a dataset
(B) To display the range of a dataset
(C) To visualize the distribution of a dataset
(D) To calculate the standard deviation
(E) To find the median of a dataset
- Q2.** Which part of a box-and-whisker plot represents the interquartile range?
(A) The whiskers
(B) The box
(C) The median line
(D) The outliers
(E) The axes
- Q3.** What is the formula for the interquartile range (IQR)?
(A) $IQR = Q_3 - Q_1$
(B) $IQR = Q_1 - Q_3$
(C) $IQR = \frac{Q_3 + Q_1}{2}$
(D) $IQR = \frac{Q_3 - Q_1}{2}$
(E) $IQR = Q_3 \cdot Q_1$
- Q4.** Which of the following is a characteristic of an outlier in a box-and-whisker plot?
(A) It is within $1.5 \cdot IQR$ of Q_1 or Q_3
(B) It is more than $1.5 \cdot IQR$ away from Q_1 or Q_3
(C) It is the median of the dataset
(D) It is the mean of the dataset
(E) It is the mode of the dataset
- Q5.** What does the median line in a box-and-whisker plot represent?
(A) The range of the dataset
(B) The interquartile range
(C) The mean of the dataset
(D) The second quartile (Q_2)
(E) The first quartile (Q_1)
- Q6.** Which of the following datasets has the largest interquartile range?
(A) 1, 2, 3, 4, 5
(B) 1, 2, 3, 4, 10
(C) 1, 2, 3, 4, 100
(D) 10, 20, 30, 40, 50
(E) 100, 200, 300, 400, 500
- Q7.** What is the first quartile (Q_1) of the dataset 1, 2, 3, 4, 5, 6, 7, 8, 9?
(A) 2
(B) 3
(C) 4
(D) 5
(E) 6
- Q8.** Which of the following statements about box-and-whisker plots is false?
(A) The plot displays the five-number summary
(B) The plot shows the distribution of the data
(C) The plot is used to compare multiple datasets
(D) The plot is only used for small datasets
(E) The plot can be used to identify outliers

Open Ended Questions (FRQ)

- Q9.** Construct a box-and-whisker plot for the dataset 12, 15, 18, 20, 22, 25, 30, 35, 40, 45, 50. Identify the median, quartiles, and any outliers.
- Q10.** Compare the box-and-whisker plots of the datasets 1, 2, 3, 4, 5 and 1, 2, 3, 4, 10. How do the plots differ? What do these differences indicate about the datasets?

Chef's Tips

- Tip 1: Emphasize the importance of correctly identifying quartiles and the interquartile range
- Tip 2: Remind students to consider the context of the data when interpreting box-and-whisker plots
- Tip 3: Encourage students to use technology to construct and compare box-and-whisker plots